

Reposys Campus Reprography Automation System

Software Engineering System Design & Proposal Documentation

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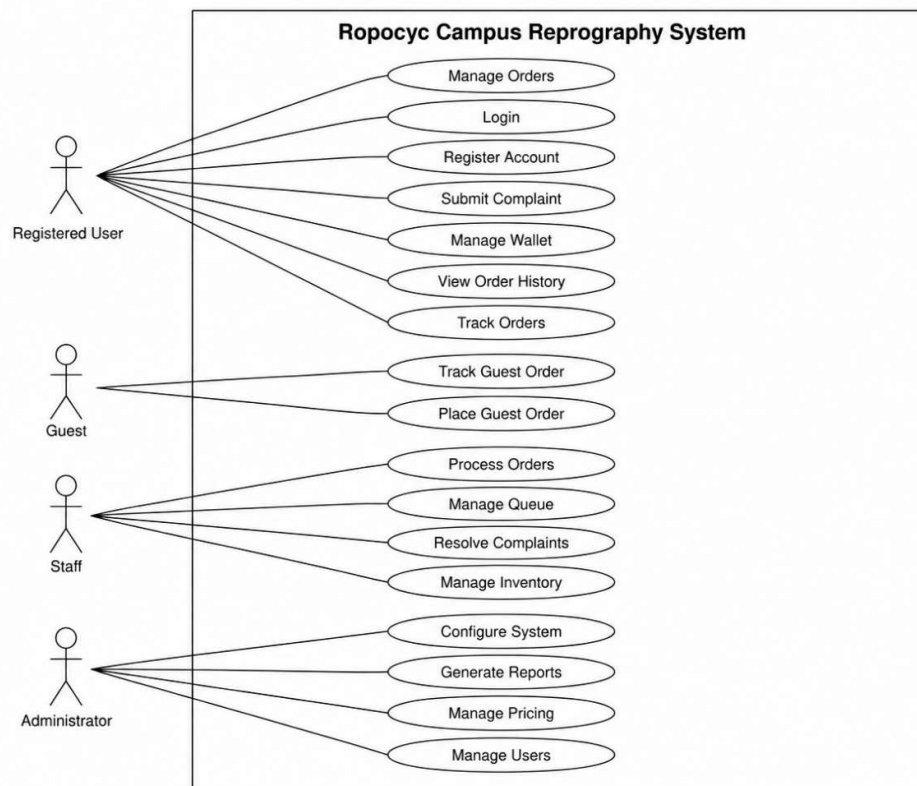
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1. System Architecture & UML Diagrams

This section provides a rigorous design blueprint of the proposed system's architecture using standard Unified Modeling Language (UML) notations. These diagrams act as the core structural foundation for the upcoming development phase.

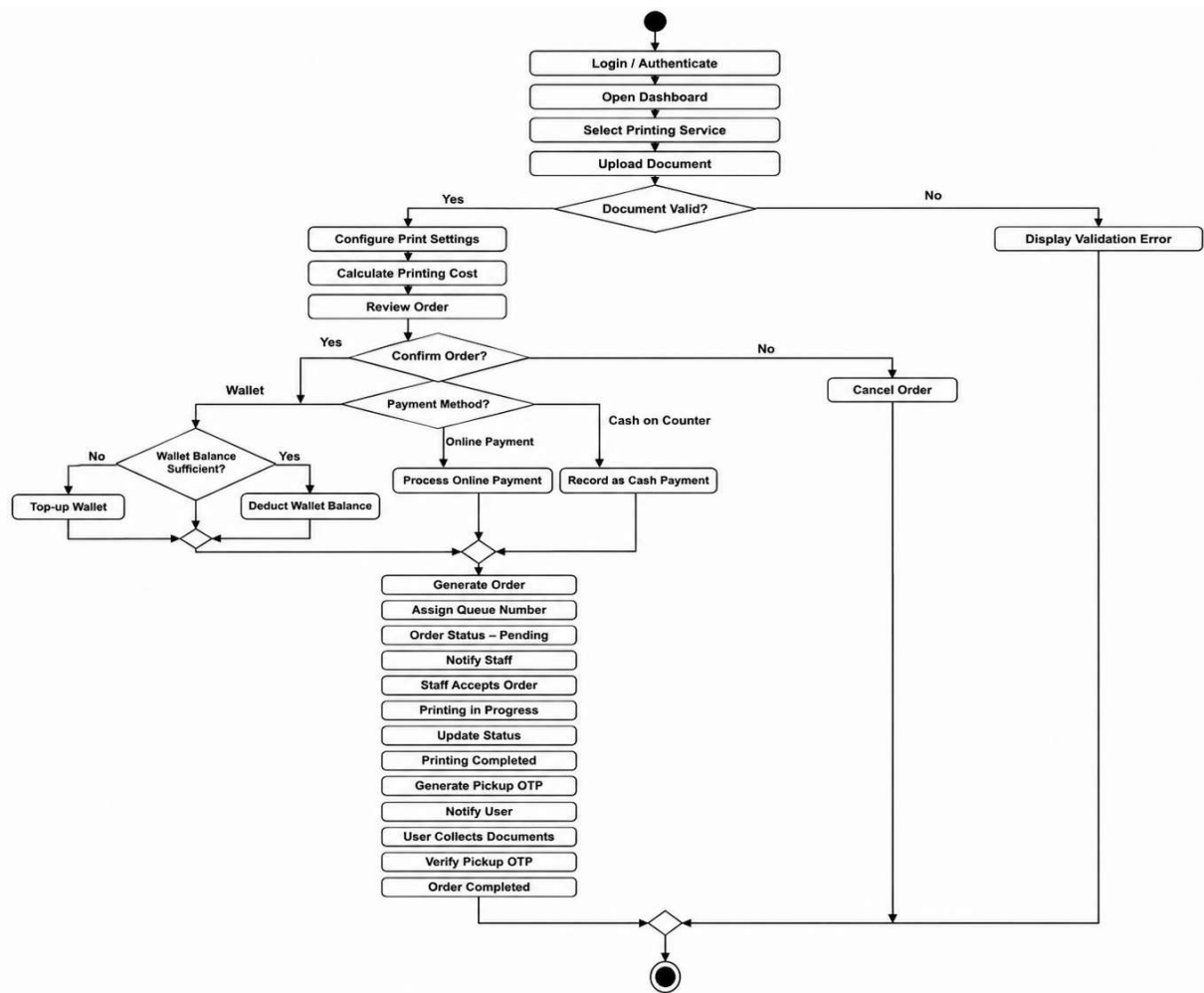
1.1 Use Case Diagram

The Use Case diagram identifies the primary actors (Registered Users, Guests, Staff, Admin) and the various system functionalities they will interact with. Students and Teachers share the exact same functional privileges within the system and are collectively modeled simply as "Users".

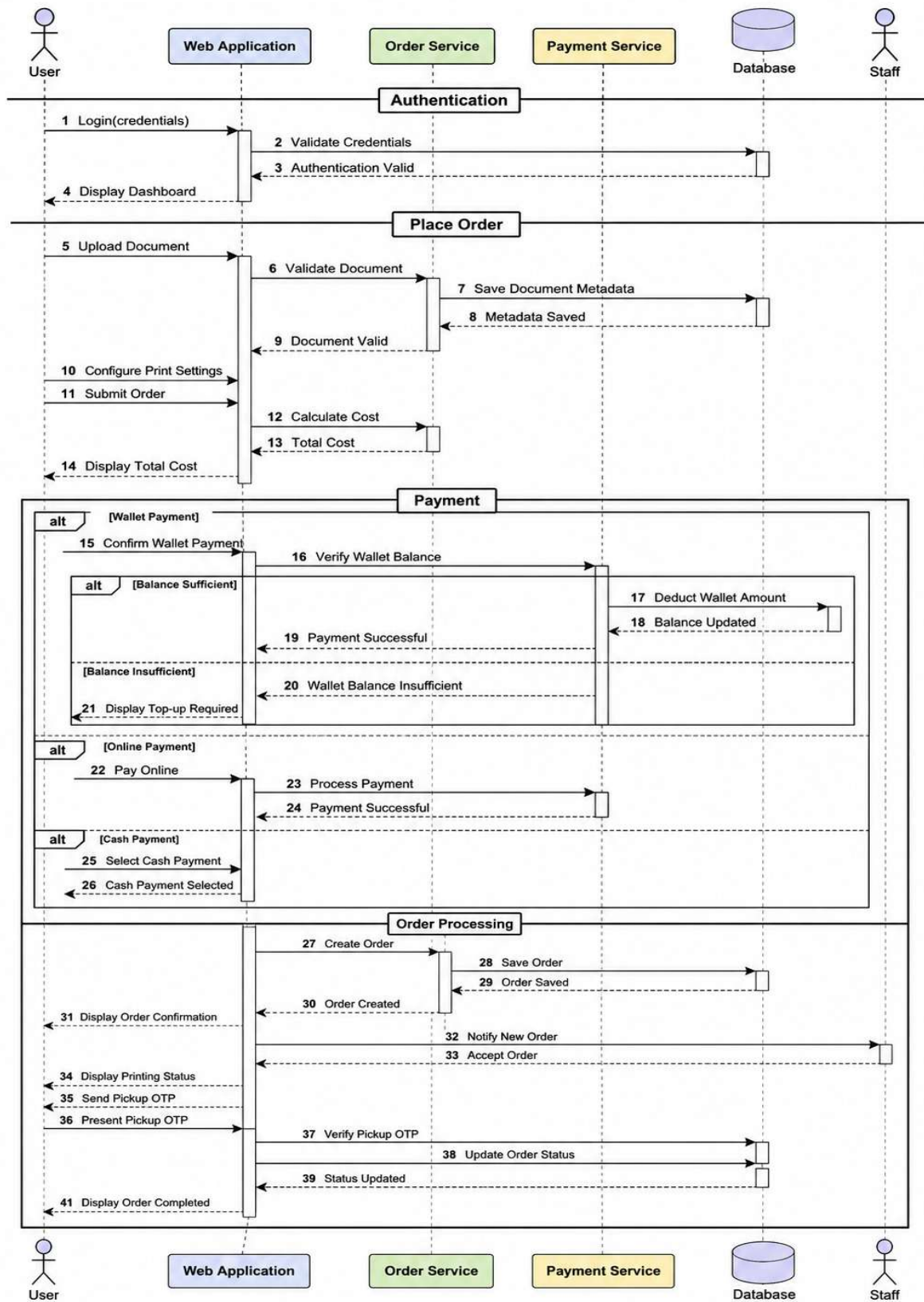


1.2 Activity Diagram: Proposed Document Print Workflow

This activity diagram illustrates the intended sequential flow of a student uploading a document, configuring the print settings, completing the digital payment, and tracking the job status.



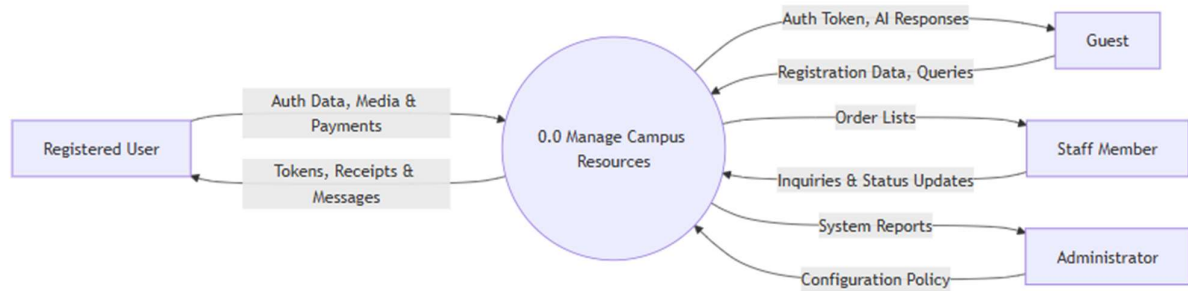
1.3 Sequence Diagram



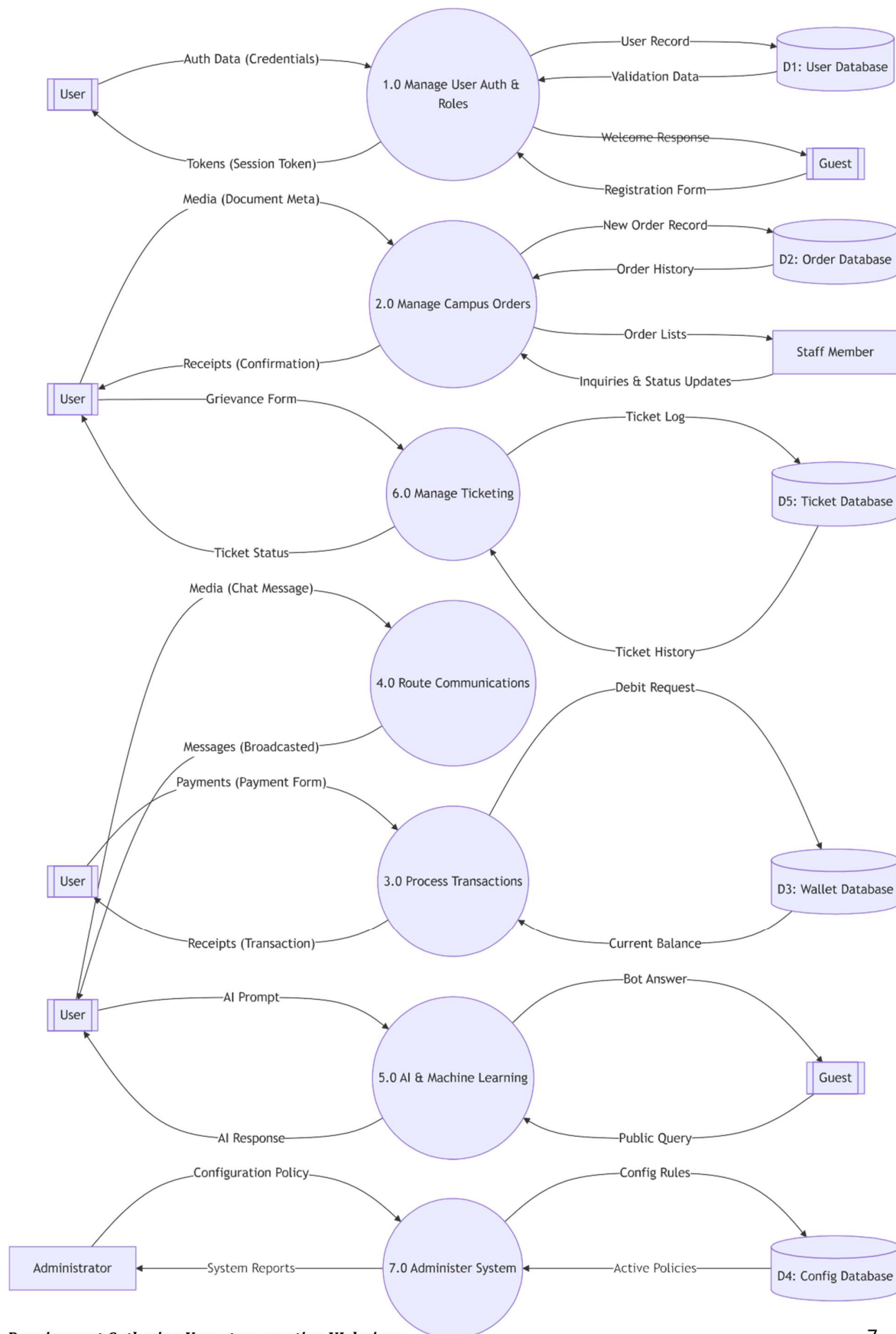
1.4 Data Flow Diagrams

Level 0 Context Diagram

The Level 0 Context Diagram defines the external entities interacting with the proposed Reposys boundary.

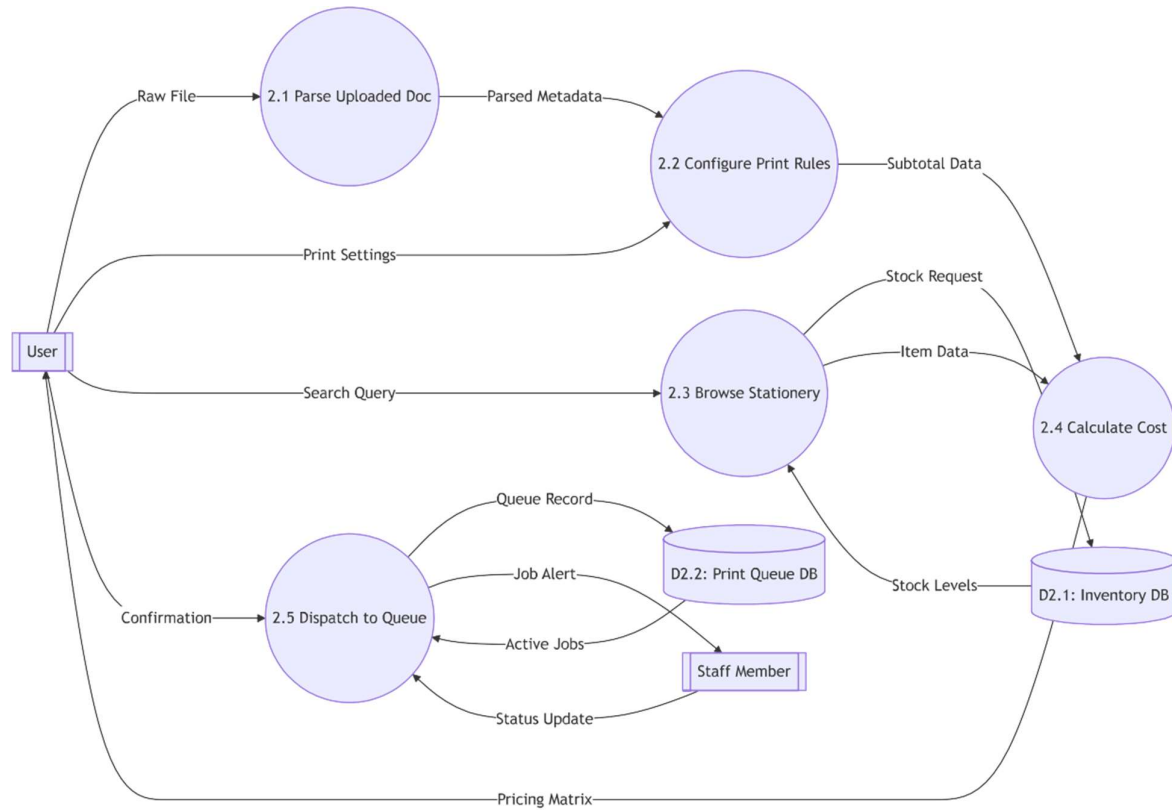


Level 1 Data Flow Diagram (Main Systems)

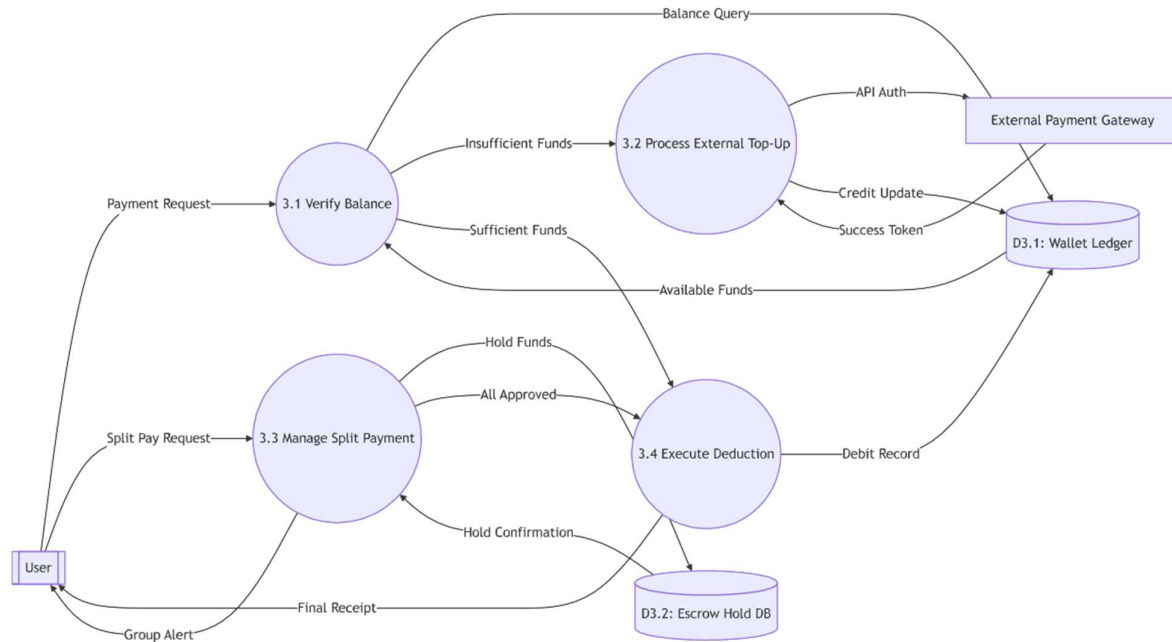


Level 2 Data Flow Diagram: 2.0 Manage Campus Orders

This Level 2 DFD explodes process 2.0 to demonstrate the granular data flows for Document Parsing, Printing, Stationery Inventory, and Cart management.

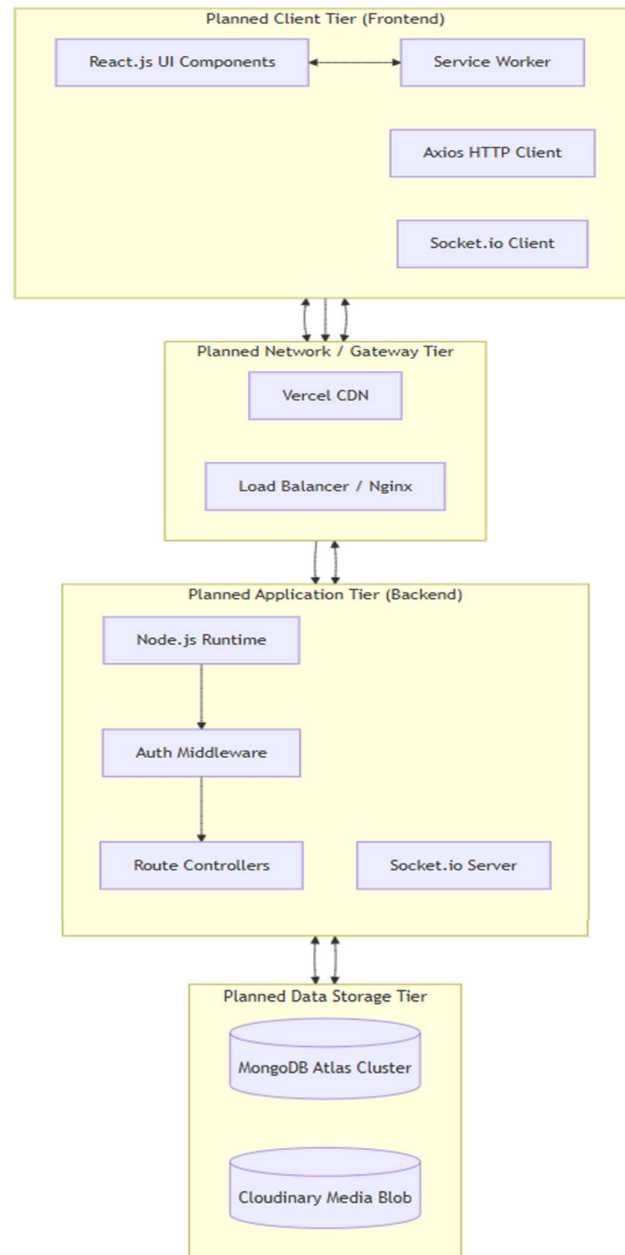


Level 2 Data Flow Diagram: 3.0 Process Transactions



2.5 Component & Deployment Diagram

This diagram represents the planned cloud-native deployment strategy, illustrating how client applications will communicate with the backend services through an API Gateway and load balancer.



2. API & Backend Route Definitions

The backend architecture will expose a RESTful API wrapped with custom JWT authentication middleware, strict IP-based rate limiting, and role-based access control models. The following endpoints will form the backbone of client-server data exchange.

Planned API Architecture Standards

- **Status Codes:** The system will adhere to standard REST HTTP codes: 200 (OK), 201 (Created), 400 (Bad Request), 401 (Unauthorized), 403 (Forbidden), 404 (Not Found), and 500 (Internal Server Error).
- **Response Envelope:** All APIs will be required to return a uniform JSON format: { "success": boolean, "data": Object/Array, "message": string }.

Middleware Stack: Every incoming request will systematically pass through Helmet (for Security Headers), CORS origin verifiers, Express JSON parser, Rate Limiter, and finally JWT token verifiers before hitting the controller logic.

Planned HTTP Method	Endpoint Route (URI)	Mapped Controller	Auth Requirement
GET	/api/admin/dashboard-stats	admin.js	Secured (JWT Required)
GET	/api/admin/pricing	admin.js	Secured (JWT Required)
PATCH	/api/admin/pricing	admin.js	Secured (JWT Required)
PATCH	/api/admin/shop/toggle	admin.js	Secured (JWT)

POST	/api/admin/follow-schedule	admin.js	Secured (JWT Required)
PATCH	/api/admin/queue/reorder	admin.js	Secured (JWT Required)
GET	/api/admin/activity-logs	admin.js	Secured (JWT Required)
DELETE	/api/admin/activity-logs	admin.js	Secured (JWT Required)
GET	/api/admin/users	admin.js	Secured (JWT Required)
POST	/api/admin/users/create-staff	admin.js	Secured (JWT Required)
PATCH	/api/admin/users/:id/toggle-active	admin.js	Secured (JWT Required)
PATCH	/api/admin/users/:id/limit-override	admin.js	Secured (JWT Required)
GET	/api/admin/users/:id	admin.js	Secured (JWT Required)
GET	/api/admin/transactions	admin.js	Secured (JWT Required)
GET	/api/admin/group-orders/:orderId/breakdown	admin.js	Secured (JWT Required)

GET	/api/admin/ratings/summary	admin.js	Secured (JWT Required)
GET	/api/admin/inquiries	admin.js	Secured (JWT Required)
PATCH	/api/admin/inquiries/:id/status	admin.js	Secured (JWT Required)
PATCH	/api/admin/inquiries/:id/respond	admin.js	Secured (JWT Required)
DELETE	/api/admin/inquiries/:id	admin.js	Secured (JWT Required)
POST	/api/admin/staff	admin.js	Secured (JWT Required)
DELETE	/api/admin/staff/:id	admin.js	Secured (JWT Required)
PATCH	/api/admin/users/:id/approve-role	admin.js	Secured (JWT Required)
PATCH	/api/admin/users/:id/reject-role	admin.js	Secured (JWT Required)
GET	/api/admin/staff/assignments	admin.js	Secured (JWT Required)
PATCH	/api/admin/users/:id/queue-assignment	admin.js	Secured (JWT Required)

GET	/api/admin/users/:id	admin.js	Secured (JWT Required)
PUT	/api/admin/users/:id	admin.js	Secured (JWT Required)
DELETE	/api/admin/users/:id	admin.js	Secured (JWT Required)
POST	/api/auth/register	auth.js	Secured (JWT Required)
GET	/api/auth/verify-email	auth.js	Secured (JWT Required)
POST	/api/auth/login	auth.js	Secured (JWT Required)
POST	/api/auth/resend-verification	auth.js	Secured (JWT Required)
POST	/api/auth/logout	auth.js	Secured (JWT Required)
POST	/api/auth/forgot-password	auth.js	Secured (JWT Required)
POST	/api/auth/reset-password	auth.js	Secured (JWT Required)
GET	/api/auth/me	auth.js	Secured (JWT Required)

GET	/api/auth/admin-test	auth.js	Secured (JWT Required)
POST	/api/chatbot/ask	chatbot.js	Secured (JWT Required)
POST	/api/complaints/	complaints.js	Secured (JWT Required)
POST	/api/complaints/:id/messages	complaints.js	Secured (JWT Required)
PATCH	/api/complaints/:id/status	complaints.js	Secured (JWT Required)
GET	/api/complaints/	complaints.js	Secured (JWT Required)
GET	/api/complaints/:id	complaints.js	Secured (JWT Required)
GET	/api/config/shop-status	config.js	Public Access
GET	/api/config/limits	config.js	Public Access
GET	/api/config/push	config.js	Public Access
POST	/api/contact/	contact.js	Secured (JWT Required)
GET	/api/contact/	contact.js	Secured (JWT Required)

PATCH	/api/contact/:id/status	contact.js	Secured (JWT Required)
PATCH	/api/contact/:id/respond	contact.js	Secured (JWT Required)
DELETE	/api/contact/:id	contact.js	Secured (JWT Required)
POST	/api/convertRoutes/docx	convertRoutes.js	Secured (JWT Required)
POST	/api/documents/upload	documents.js	Secured (JWT Required)
POST	/api/documents/upload-multiple	documents.js	Secured (JWT Required)
GET	/api/documents/library	documents.js	Secured (JWT Required)
GET	/api/documents/:id/url	documents.js	Secured (JWT Required)
GET	/api/friendshipRoutes/search	friendshipRoutes.js	Secured (JWT Required)
POST	/api/friendshipRoutes/request	friendshipRoutes.js	Secured (JWT Required)
POST	/api/friendshipRoutes/respond	friendshipRoutes.js	Secured (JWT Required)

GET	/api/friendshipRoutes/	friendshipRoute s.js	Secured (JWT Required)
GET	/api/friendshipRoutes/pending	friendshipRoute s.js	Secured (JWT Required)
DELETE	/api/friendshipRoutes/:friendsh ipId	friendshipRoute s.js	Secured (JWT Required)
POST	/api/groupChatRoutes/	groupChatRoute s.js	Secured (JWT Required)
GET	/api/groupChatRoutes/	groupChatRoute s.js	Secured (JWT Required)
GET	/api/groupChatRoutes/:groupId /messages	groupChatRoute s.js	Secured (JWT Required)
POST	/api/groupChatRoutes/:groupId /messages	groupChatRoute s.js	Secured (JWT Required)
POST	/api/groupChatRoutes/:groupId /send-voice	groupChatRoute s.js	Secured (JWT Required)
POST	/api/groupChatRoutes/:groupId /add-member	groupChatRoute s.js	Secured (JWT Required)
DELETE	/api/groupChatRoutes/:groupId /remove-member	groupChatRoute s.js	Secured (JWT Required)
POST	/api/groupChatRoutes/:groupId /leave	groupChatRoute s.js	Secured (JWT Required)

POST	/api/groupChatRoutes/:groupId /disband	groupChatRoute s.js	Secured (JWT Required)
POST	/api/groupChatRoutes/:groupId /delete-chat	groupChatRoute s.js	Secured (JWT Required)
GET	/api/groupOrderRoutes/	groupOrderRoute es.js	Secured (JWT Required)
GET	/api/groupOrderRoutes/my- active	groupOrderRoute es.js	Secured (JWT Required)
GET	/api/groupOrderRoutes/group- chat/:groupId	groupOrderRoute es.js	Secured (JWT Required)
POST	/api/groupOrderRoutes/custom- split	groupOrderRoute es.js	Secured (JWT Required)
POST	/api/groupOrderRoutes/split/cu stom	groupOrderRoute es.js	Secured (JWT Required)
POST	/api/groupOrderRoutes/split/se nd	groupOrderRoute es.js	Secured (JWT Required)
POST	/api/groupOrderRoutes/split/re spond	groupOrderRoute es.js	Secured (JWT Required)
GET	/api/groupOrderRoutes/split/pe nding	groupOrderRoute es.js	Secured (JWT Required)
GET	/api/groupOrderRoutes/:orderId	groupOrderRoute es.js	Secured (JWT Required)

GET	/api/groupOrderRoutes/:group OrderId/split-status	groupOrderRout es.js	Secured (JWT Required)
POST	/api/guest/request-otp	guest.js	Secured (JWT Required)
POST	/api/guest/verify-otp	guest.js	Secured (JWT Required)
POST	/api/guest/end-session	guest.js	Secured (JWT Required)
GET	/api/inventory/	inventory.js	Secured (JWT Required)
GET	/api/inventory/summary	inventory.js	Secured (JWT Required)
PATCH	/api/inventory/:id/update	inventory.js	Secured (JWT Required)
POST	/api/messageRoutes/send	messageRoutes.j s	Secured (JWT Required)
POST	/api/messageRoutes/send- voice/:receiverId	messageRoutes.j s	Secured (JWT Required)
GET	/api/messageRoutes/unread- count	messageRoutes.j s	Secured (JWT Required)
GET	/api/messageRoutes/conversatio ns	messageRoutes.j s	Secured (JWT Required)

GET	/api/messageRoutes/:friendId	messageRoutes.js	Secured (JWT Required)
PUT	/api/messageRoutes/:friendId/read	messageRoutes.js	Secured (JWT Required)
DELETE	/api/messageRoutes/:messageId	messageRoutes.js	Secured (JWT Required)
GET	/api/notifications/	notifications.js	Secured (JWT Required)
GET	/api/notifications/all	notifications.js	Secured (JWT Required)
GET	/api/notifications/unread-count	notifications.js	Secured (JWT Required)
PATCH	/api/notifications/:id/read	notifications.js	Secured (JWT Required)
PATCH	/api/notifications/read-all	notifications.js	Secured (JWT Required)
DELETE	/api/notifications/:id	notifications.js	Secured (JWT Required)
POST	/api/orders/estimate	orders.js	Secured (JWT Required)
GET	/api/orders/track/:orderId	orders.js	Secured (JWT Required)

GET	/api/orders/my-orders	orders.js	Secured (JWT Required)
POST	/api/orders/create	orders.js	Secured (JWT Required)
POST	/api/orders/:id/reorder	orders.js	Secured (JWT Required)
GET	/api/orders/:id	orders.js	Secured (JWT Required)
GET	/api/orders/:id/wait	orders.js	Secured (JWT Required)
GET	/api/orders/:id/receipt	orders.js	Secured (JWT Required)
PATCH	/api/orders/:id/cancel	orders.js	Secured (JWT Required)
POST	/api/payments/webhook	payments.js	Secured (JWT Required)
POST	/api/payments/create-order	payments.js	Secured (JWT Required)
POST	/api/payments/verify	payments.js	Secured (JWT Required)
POST	/api/payments/cash-confirm	payments.js	Secured (JWT Required)

POST	/api/ratings/	ratings.js	Secured (JWT Required)
GET	/api/ratings/my-ratings	ratings.js	Secured (JWT Required)
GET	/api/receiptRoutes/:orderId	receiptRoutes.js	Secured (JWT Required)
GET	/api/reports/daily-orders	reports.js	Secured (JWT Required)
GET	/api/reports/revenue	reports.js	Secured (JWT Required)
GET	/api/reports/service-breakdown	reports.js	Secured (JWT Required)
GET	/api/reports/peak-hours	reports.js	Secured (JWT Required)
GET	/api/reports/staff-performance	reports.js	Secured (JWT Required)
GET	/api/reports/payment-breakdown	reports.js	Secured (JWT Required)
GET	/api/reports/export	reports.js	Secured (JWT Required)
GET	/api/staff/queue	staff.js	Secured (JWT Required)

GET	/api/staff/staff/orders/complete d	staff.js	Secured (JWT Required)
PATCH	/api/staff/orders/:id/lock	staff.js	Secured (JWT Required)
PATCH	/api/staff/orders/:id/unlock	staff.js	Secured (JWT Required)
POST	/api/staff/orders/:id/start- processing	staff.js	Secured (JWT Required)
POST	/api/staff/orders/:id/ready-for- pickup	staff.js	Secured (JWT Required)
POST	/api/staff/orders/:id/collect- cash	staff.js	Secured (JWT Required)
POST	/api/staff/orders/:id/verify-otp	staff.js	Secured (JWT Required)
POST	/api/staff/orders/:id/partial	staff.js	Secured (JWT Required)
POST	/api/staffReports/	staffReports.js	Secured (JWT Required)
GET	/api/staffReports/	staffReports.js	Secured (JWT Required)
PATCH	/api/staffReports/:id/acknowled ge	staffReports.js	Secured (JWT Required)

PATCH	/api/staffReports/:id/resolve	staffReports.js	Secured (JWT Required)
POST	/api/tools/convert	tools.js	Secured (JWT Required)
POST	/api/tools/resize-image	tools.js	Secured (JWT Required)
POST	/api/tools/compress-image	tools.js	Secured (JWT Required)
POST	/api/tools/resize-pdf-pages	tools.js	Secured (JWT Required)
POST	/api/uploadRoutes/	uploadRoutes.js	Secured (JWT Required)
GET	/api/users/me	users.js	Secured (JWT Required)
PATCH	/api/users/me	users.js	Secured (JWT Required)
PATCH	/api/users/me/change-password	users.js	Secured (JWT Required)
POST	/api/users/me/logout-all-devices	users.js	Secured (JWT Required)
GET	/api/users/me/usage-summary	users.js	Secured (JWT Required)

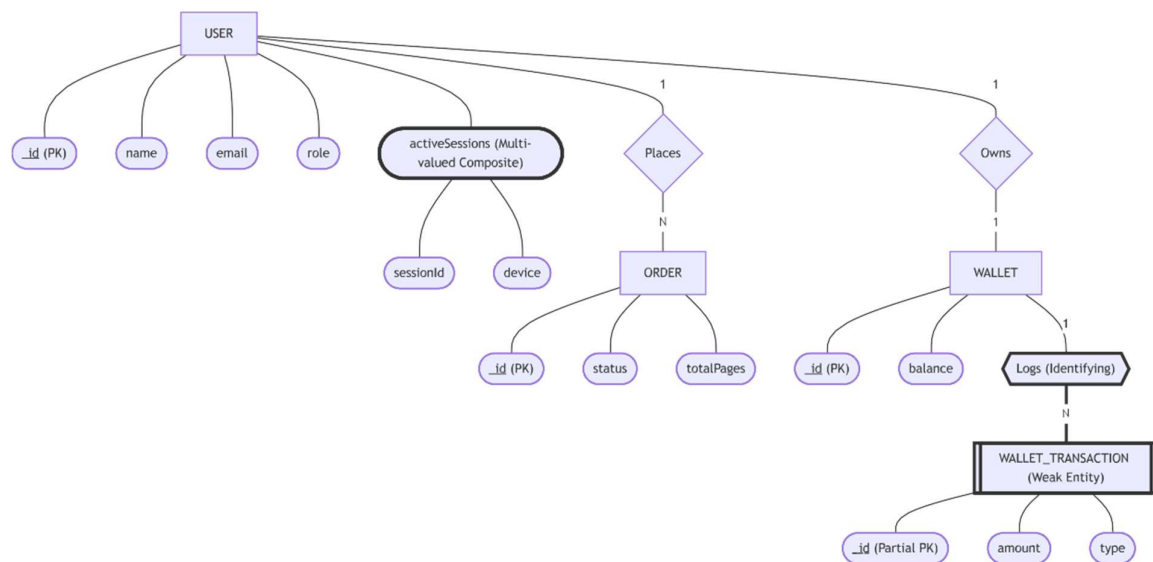
POST	/api/users/push-subscription	users.js	Secured (JWT Required)
POST	/api/users/test-push	users.js	Secured (JWT Required)
DELETE	/api/users/push-subscription	users.js	Secured (JWT Required)
GET	/api/walletRoutes/	walletRoutes.js	Secured (JWT Required)
GET	/api/walletRoutes/transactions	walletRoutes.js	Secured (JWT Required)
POST	/api/walletRoutes/topup/create-order	walletRoutes.js	Secured (JWT Required)
POST	/api/walletRoutes/topup/verify	walletRoutes.js	Secured (JWT Required)
POST	/api/walletRoutes/complete-onboarding	walletRoutes.js	Secured (JWT Required)

3. Database Design & Data Models

The system is designed to utilize a NoSQL document-based architecture via MongoDB. Data integrity will be maintained securely using Mongoose schemas. Below is the proposed structural design of the core database entities.

Complete Entity-Relationship (ER) Conceptual Model (Chen's Notation)

This ER model strictly adheres to conceptual design rules: independent entities (rectangles), attributes (ovals), active relationships (diamonds), and correctly mapped structural constraints. It accurately reflects the actual MongoDB schema without injecting fake constraints.



3.ActivityLog Planned Schema: **ActivityLog**

Proposed Field Name	Data Type / Constraints	Expected Behavior
actionType	String	Required
performedBy	ObjectId (Foreign Key)	Optional
description	String	Required
affectedRecordId	ObjectId (Foreign Key)	Optional
metadata	Mixed Object	Optional
timestamp	Date	Optional Default: Date.now
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3.Complaint Planned Schema: Complaint

Proposed Field Name	Data Type / Constraints	Expected Behavior
status	String	Required
timestamp	Date	Optional Default: Date.now
actorId	ObjectId (Foreign Key)	Optional Default: null
actorRole	String	Optional Default: null
note	String	Optional Default: "
senderId	ObjectId (Foreign Key)	Optional Default: null
senderRole	String	Optional Default: null
text	String	Required
attachmentUrl	String	Optional Default: "
timestamp	Date	Optional Default: Date.now
complaintToken	String	Required
orderId	ObjectId (Foreign Key)	Required
raisedBy	ObjectId (Foreign Key)	Required
category	String	Required
description	String	Required
status	String	Optional Default: 'Open'
statusHistory	Mixed Object	Optional Default: []
messages	Mixed Object	Optional Default: []
attachmentUrls	Mixed Object	Optional Default: []

resolvedAt	Date	Optional Default: null
type	String	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration

3.DailyCounter Planned Schema: DailyCounter

Proposed Field Name	Data Type / Constraints	Expected Behavior
date	String	Required
prefix	String	Required Default: 'Order'
count	Number	Optional Default: 0
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Number	Base Property Configuration

3.Document Planned Schema: Document

Proposed Field Name	Data Type / Constraints	Expected Behavior
page	Number	Optional
issue	String	Optional
blankPages	Mixed Object	Optional Default: []
qualityIssues	Mixed Object	Optional Default: []
isPasswordProtected	Boolean	Optional Default: false
colourHeavyPages	Mixed Object	Optional Default: []
cloudinaryUrl	String	Required
publicId	String	Required
originalFilename	String	Required
fileType	String	Required
pageCount	Number	Optional
analysisResults	Mixed Object	Optional Default: () => ({
ownerId	ObjectId (Foreign Key)	Optional
isGuest	Boolean	Optional Default: false
guestSessionId	String	Optional
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	Boolean	Base Property Configuration
type	String	Base Property Configuration

3.Friendship Planned Schema: **Friendship**

Proposed Field Name	Data Type / Constraints	Expected Behavior
requester	ObjectId (Foreign Key)	Required
recipient	ObjectId (Foreign Key)	Required
status	String	Optional Default: 'pending'
createdAt	Date	Optional Default: Date.now
updatedAt	Date	Optional Default: Date.now
type	String	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3.Group Planned Schema: **Group**

Proposed Field Name	Data Type / Constraints	Expected Behavior
name	String	Required
createdBy	ObjectId (Foreign Key)	Required
members	ObjectId (Foreign Key)	Optional
validate	Mixed Object	Optional
isDisbanded	Boolean	Optional Default: false
createdAt	Date	Optional Default: Date.now
type	String	Base Property Configuration
type	Boolean	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3.GroupOrder Planned Schema: GroupOrder

Proposed Field Name	Data Type / Constraints	Expected Behavior
userId	ObjectId (Foreign Key)	Required
amount	Number	Required
walletStatus	String	Optional Default: 'pending'
paidAt	Date	Optional Default: null
orderId	ObjectId (Foreign Key)	Optional Default: null
createdBy	ObjectId (Foreign Key)	Required
totalAmount	Number	Required
participants	Mixed Object	Required
description	String	Optional Default: "
status	String	Optional Default: 'active'
createdAt	Date	Optional Default: Date.now
partialFilterExpression	Mixed Object	Optional
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3.GuestSession Planned Schema: GuestSession

Proposed Field Name	Data Type / Constraints	Expected Behavior
email	String	Required
hashedOtp	String	Optional
otpExpiry	Date	Optional
otpRequestCount	Number	Optional Default: 0
otpRequestWindowStart	Date	Optional
sessionId	String	Optional Default: uuid
sessionToken	String	Optional
isActive	Boolean	Optional Default: true
createdAt	Date	Optional Default: Date.now
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
type	Number	Base Property Configuration
type	Date	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Boolean	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3. Inquiry Planned Schema: Inquiry

Proposed Field Name	Data Type / Constraints	Expected Behavior
name	String	Optional
email	String	Optional
subject	String	Optional
message	String	Optional
status	String	Optional Default: 'Pending'
response	String	Optional
respondedAt	Date	Optional
respondedBy	ObjectId (Foreign Key)	Optional
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration

3.InventoryItem Planned Schema: `InventoryItem`

Proposed Field Name	Data Type / Constraints	Expected Behavior
<code>name</code>	String	Required
<code>category</code>	String	Required
<code>unit</code>	String	Required
<code>currentStock</code>	Number	Required
<code>minimumThreshold</code>	Number	Required
<code>usageRate</code>	Number	Optional Default: 0
<code>daysOfStockRemaining</code>	Number	Optional Default: 0
<code>lastUpdatedBy</code>	ObjectId (Foreign Key)	Optional Default: null
<code>lastUpdatedAt</code>	Date	Optional Default: null
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Number	Base Property Configuration
<code>type</code>	Number	Base Property Configuration
<code>type</code>	Number	Base Property Configuration
<code>type</code>	Number	Base Property Configuration
<code>type</code>	Date	Base Property Configuration

3.Message Planned Schema: **Message**

Proposed Field Name	Data Type / Constraints	Expected Behavior
senderId	ObjectId (Foreign Key)	Required
recipientId	ObjectId (Foreign Key)	Optional Default: null
groupId	ObjectId (Foreign Key)	Optional Default: null
content	String	Optional Default: ""
mediaUrl	String	Optional Default: null
mediaType	String	Optional Default: 'none'
read	Boolean	Optional Default: false
type	String	Optional Default: 'text'
audioUrl	String	Optional Default: null
audioDuration	Number	Optional Default: null
splitCardData	Mixed Object	Optional Default: null
isSystem	Boolean	Optional Default: false
isDeleted	Boolean	Optional Default: false
timestamp	Date	Optional Default: Date.now
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Boolean	Base Property Configuration

type	String	Base Property Configuration
type	String	Base Property Configuration
type	Number	Base Property Configuration
type	Boolean	Base Property Configuration
type	Boolean	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3.Notification Planned Schema: Notification

Proposed Field Name	Data Type / Constraints	Expected Behavior
recipientId	ObjectId (Foreign Key)	Optional Default: null
recipientRole	String	Optional Default: null
title	String	Required
message	String	Required
type	String	Required Default: 'system'
recipientType	String	Required
urgency	String	Optional Default: 'Normal'
relatedEntity	String	Optional
isRead	Boolean	Optional Default: false
relatedOrderId	ObjectId (Foreign Key)	Optional Default: null
relatedServiceType	String	Optional Default: null
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Boolean	Base Property Configuration
type	String	Base Property Configuration

3.Order Planned Schema: Order

Proposed Field Name	Data Type / Constraints	Expected Behavior
status	String	Optional
actorId	ObjectId (Foreign Key)	Optional
actorRole	String	Optional
timestamp	Date	Optional Default: Date.now
note	String	Optional
copies	Number	Optional Default: 1
colourMode	String	Optional Default: 'BlackAndWhite'
sided	String	Optional Default: 'Single'
paperSize	String	Optional Default: 'A4'
binding	String	Optional Default: 'None'
printInstructions	String	Optional
outputFormat	String	Optional
conversionType	String	Optional
userId	ObjectId (Foreign Key)	Optional
isGuest	Boolean	Optional Default: false
guestEmail	String	Optional
guestSessionId	String	Optional
serviceType	String	Required
userRole	String	Required
printConfig	Mixed Object	Optional Default: {} => ({

pageCount	Number	Optional
estimatedCost	Number	Optional
finalCost	Number	Optional
paymentMethod	String	Optional
paymentStatus	String	Optional Default: 'Pending'
razorpayOrderId	String	Optional
razorpayPaymentId	String	Optional
status	String	Optional Default: 'Pending'
statusHistory	Mixed Object	Optional Default: []
tokenNumber	String	Optional
priorityScore	Number	Optional Default: 0
agingScore	Number	Optional Default: 0
preferredPickupSlot	String	Optional
assignedStaffId	ObjectId (Foreign Key)	Optional
pickupOtp	String	Optional
pickupOtpExpiry	Date	Optional
pickupOtpVerified	Boolean	Optional Default: false
otpAttempts	Number	Optional Default: 0
otpLocked	Boolean	Optional Default: false
lockedBy	ObjectId (Foreign Key)	Optional
lockedAt	Date	Optional
partialPagesCompleted	Number	Optional Default: 0

<code>followUpOrderId</code>	ObjectId (Foreign Key)	Optional
<code>internalNotes</code>	String	Optional Default: "
<code>estimatedDuration</code>	Number	Optional
<code>cancelReason</code>	String	Optional
<code>isGroupOrder</code>	Boolean	Optional Default: false
<code>groupOrderId</code>	ObjectId (Foreign Key)	Optional Default: null
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Date	Base Property Configuration
<code>default</code>	Date	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Number	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Boolean	Base Property Configuration
<code>type</code>	String	Base Property Configuration

type	String	Base Property Configuration
type	Number	Base Property Configuration
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
type	Boolean	Base Property Configuration
type	Number	Base Property Configuration
type	Boolean	Base Property Configuration
type	Date	Base Property Configuration
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	Boolean	Base Property Configuration

3.Payment Planned Schema: **Payment**

Proposed Field Name	Data Type / Constraints	Expected Behavior
orderId	ObjectId (Foreign Key)	Required
userId	ObjectId (Foreign Key)	Optional
amount	Number	Required
currency	String	Optional Default: 'INR'
razorpayOrderId	String	Optional
razorpayPaymentId	String	Optional
razorpaySignature	String	Optional
status	String	Optional Default: 'Created'
method	String	Optional
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration

3.Rating Planned Schema: Rating

Proposed Field Name	Data Type / Constraints	Expected Behavior
orderId	ObjectId (Foreign Key)	Required
userId	ObjectId (Foreign Key)	Required
serviceType	String	Required
stars	Number	Required
comment	String	Optional
createdAt	Date	Optional Default: Date.now
type	String	Base Property Configuration
type	Number	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
default	Date	Base Property Configuration

3.SplitRequest Planned Schema: `SplitRequest`

Proposed Field Name	Data Type / Constraints	Expected Behavior
<code>groupOrderId</code>	ObjectId (Foreign Key)	Required
<code>from</code>	ObjectId (Foreign Key)	Required
<code>to</code>	ObjectId (Foreign Key)	Required
<code>amount</code>	Number	Required
<code>status</code>	String	Optional Default: 'pending'
<code>triggeredFrom</code>	String	Required
<code>groupChatId</code>	ObjectId (Foreign Key)	Optional Default: null
<code>createdAt</code>	Date	Optional Default: Date.now
<code>resolvedAt</code>	Date	Optional Default: null
<code>type</code>	Number	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Date	Base Property Configuration
<code>default</code>	Date	Base Property Configuration
<code>type</code>	Date	Base Property Configuration

3.StaffReport Planned Schema: StaffReport

Proposed Field Name	Data Type / Constraints	Expected Behavior
reportRef	String	Required
raisedBy	ObjectId (Foreign Key)	Required
category	String	Required
specificItem	String	Optional
description	String	Required
urgency	String	Optional Default: 'Normal'
status	String	Optional Default: 'Pending'
resolutionNote	String	Optional
acknowledgedAt	Date	Optional
resolvedAt	Date	Optional
linkedInventoryItemId	ObjectId (Foreign Key)	Optional
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	String	Base Property Configuration
type	Date	Base Property Configuration
type	Date	Base Property Configuration

3.SystemConfig Planned Schema: SystemConfig

Proposed Field Name	Data Type / Constraints	Expected Behavior
Monday	Mixed Object	Optional
Tuesday	Mixed Object	Optional
Wednesday	Mixed Object	Optional
Thursday	Mixed Object	Optional
Friday	Mixed Object	Optional
open	String	Required
close	String	Required
pricing	Number	Optional Default: DEFAULT_PRICING.printBW
printColour	Number	Optional Default: DEFAULT_PRICING.printColour
doubleSidedMultiplier	Number	Optional Default: DEFAULT_PRICING.doubleSidedMultiplier
photocopy	Number	Optional Default: DEFAULT_PRICING.photocopy
scanning	Number	Optional Default: DEFAULT_PRICING.scanning
bindingSpiral	Number	Optional Default: DEFAULT_PRICING.bindingSpiral
bindingStaple	Number	Optional Default: DEFAULT_PRICING.bindingStaple
conversion	Number	Optional Default: DEFAULT_PRICING.conversion
printTimePerPage	Number	Optional Default: DEFAULT_PRICING.printTimePerPage
photocopyTimePerPage	Number	Optional Default: DEFAULT_PRICING.photocopyTimePerPage
scanTimePerPage	Number	Optional Default: DEFAULT_PRICING.scanTimePerPage
bindingSpiralDuration	Number	Optional Default: DEFAULT_PRICING.bindingSpiralDuration

<code>bindingStapleDuration</code>	Number	Optional Default: DEFAULT_PRICING.bindingStapleDuration
<code>conversionDuration</code>	Number	Optional Default: DEFAULT_PRICING.conversionDuration
<code>orderLimits</code>	Number	Optional Default: DEFAULT_ORDER_LIMITS.maxPagesPerOrder
<code>maxCopies</code>	Number	Optional Default: DEFAULT_ORDER_LIMITS.maxCopies
<code>maxDocumentsPerOrder</code>	Number	Optional Default: DEFAULT_ORDER_LIMITS.maxDocumentsPerOrder
<code>maxOrdersPerUserPerDay</code>	Number	Optional Default: DEFAULT_ORDER_LIMITS.maxOrdersPerUserPerDay
<code>operatingHours</code>	Mixed Object	Optional Default: () => cloneValue(DEFAULT_OPERATING_HOURS.Monday)
<code>Tuesday</code>	Mixed Object	Optional Default: () => cloneValue(DEFAULT_OPERATING_HOURS.Tuesday)
<code>Wednesday</code>	Mixed Object	Optional Default: () => cloneValue(DEFAULT_OPERATING_HOURS.Wednesday)
<code>Thursday</code>	Mixed Object	Optional Default: () => cloneValue(DEFAULT_OPERATING_HOURS.Thursday)
<code>Friday</code>	Mixed Object	Optional Default: () => cloneValue(DEFAULT_OPERATING_HOURS.Friday)
<code>Saturday</code>	Mixed Object	Optional Default: null
<code>Sunday</code>	Mixed Object	Optional Default: null
<code>isManuallyOpen</code>	Boolean	Optional Default: null
<code>shopMode</code>	String	Optional Default: 'manual_open'
<code>starvationThresholdMinutes</code>	Number	Optional Default: 60
<code>starvationBoost</code>	Number	Optional Default: 20

type	String	Base Property Configuration
type	String	Base Property Configuration
type	Number	Base Property Configuration
type	Number	Base Property Configuration
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type	Number	Base Property Configuration
type	Number	Base Property Configuration
type	Number	Base Property Configuration
type	Number	Base Property Configuration
type	Number	Base Property Configuration
type	Number	Base Property Configuration
type	Boolean	Base Property Configuration
type	String	Base Property Configuration

3. User Planned Schema: User

Proposed Field Name	Data Type / Constraints	Expected Behavior
name	String	Required
email	String	Required
password	String	Required
collegeId	String	Required
department	String	Required
role	String	Required
pendingRole	String	Optional Default: null
pendingRoleApproval	Boolean	Optional Default: false
verified	Boolean	Optional Default: false
verificationToken	String	Optional
verificationTokenExpiry	Date	Optional
pendingEmail	String	Optional
emailChangeToken	String	Optional
emailChangeTokenExpiry	Date	Optional
resetPasswordToken	String	Optional
resetPasswordExpiry	Date	Optional
isActive	Boolean	Optional Default: true
totalOrders	Number	Optional Default: 0
totalSpend	Number	Optional Default: 0
phone	String	Optional

<code>sessionInvalidatedAt</code>	Date	Optional
<code>queueAssignment</code>	String	Optional Default: 'All'
<code>pushSubscription</code>	Mixed Object	Optional Default: null
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Boolean	Base Property Configuration
<code>type</code>	Boolean	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Date	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Date	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Date	Base Property Configuration
<code>type</code>	Boolean	Base Property Configuration
<code>type</code>	Number	Base Property Configuration

sessionId	String	Base Property Configuration
device	String	Base Property Configuration
lastActive	Date	Base Property Configuration
type	Date	Base Property Configuration
service	String	Base Property Configuration
maxPages	Number	Base Property Configuration
validUntil	Date	Base Property Configuration
type	String	Base Property Configuration

3.Wallet Planned Schema: wallet

Proposed Field Name	Data Type / Constraints	Expected Behavior
userId	ObjectId (Foreign Key)	Required
balance	Number	Optional Default: 0
hasSeenOnboarding	Boolean	Optional Default: false
type	Number	Base Property Configuration
type	Boolean	Base Property Configuration

3.WalletTransaction Planned Schema: `WalletTransaction`

Proposed Field Name	Data Type / Constraints	Expected Behavior
<code>walletId</code>	ObjectId (Foreign Key)	Required
<code>type</code>	String	Required
<code>amount</code>	Number	Required
<code>orderId</code>	ObjectId (Foreign Key)	Optional Default: null
<code>razorpayOrderId</code>	String	Optional Default: null
<code>description</code>	String	Required
<code>timestamp</code>	Date	Optional Default: Date.now
<code>status</code>	String	Optional Default: 'success'
<code>type</code>	String	Base Property Configuration
<code>type</code>	Number	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	String	Base Property Configuration
<code>type</code>	Date	Base Property Configuration
<code>default</code>	Date	Base Property Configuration
<code>type</code>	String	Base Property Configuration

4. UI/UX & Frontend Design Specifications

The frontend presentation layer will be constructed using **React.js** and **Tailwind CSS**, utilizing mobile-first responsive design principles, glassmorphism aesthetics, and comprehensive Dark Mode support.

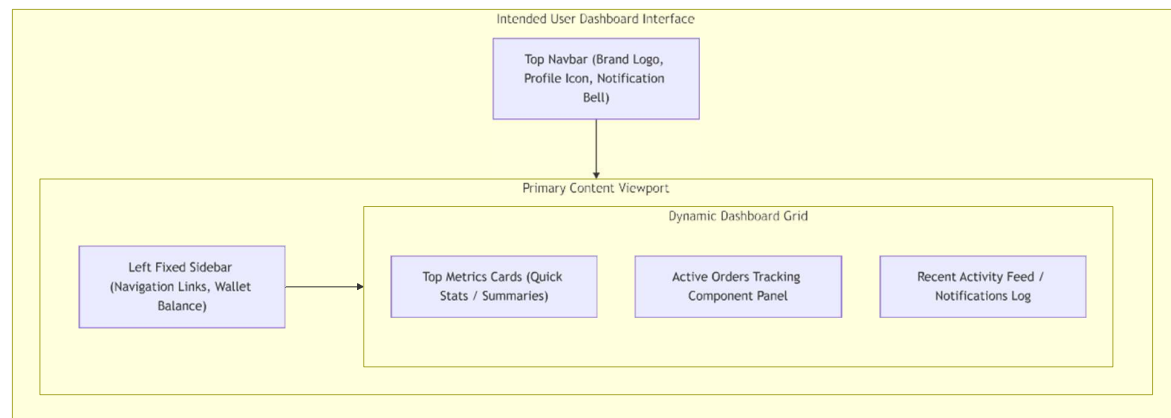
4.1 Proposed Design System & Guidelines

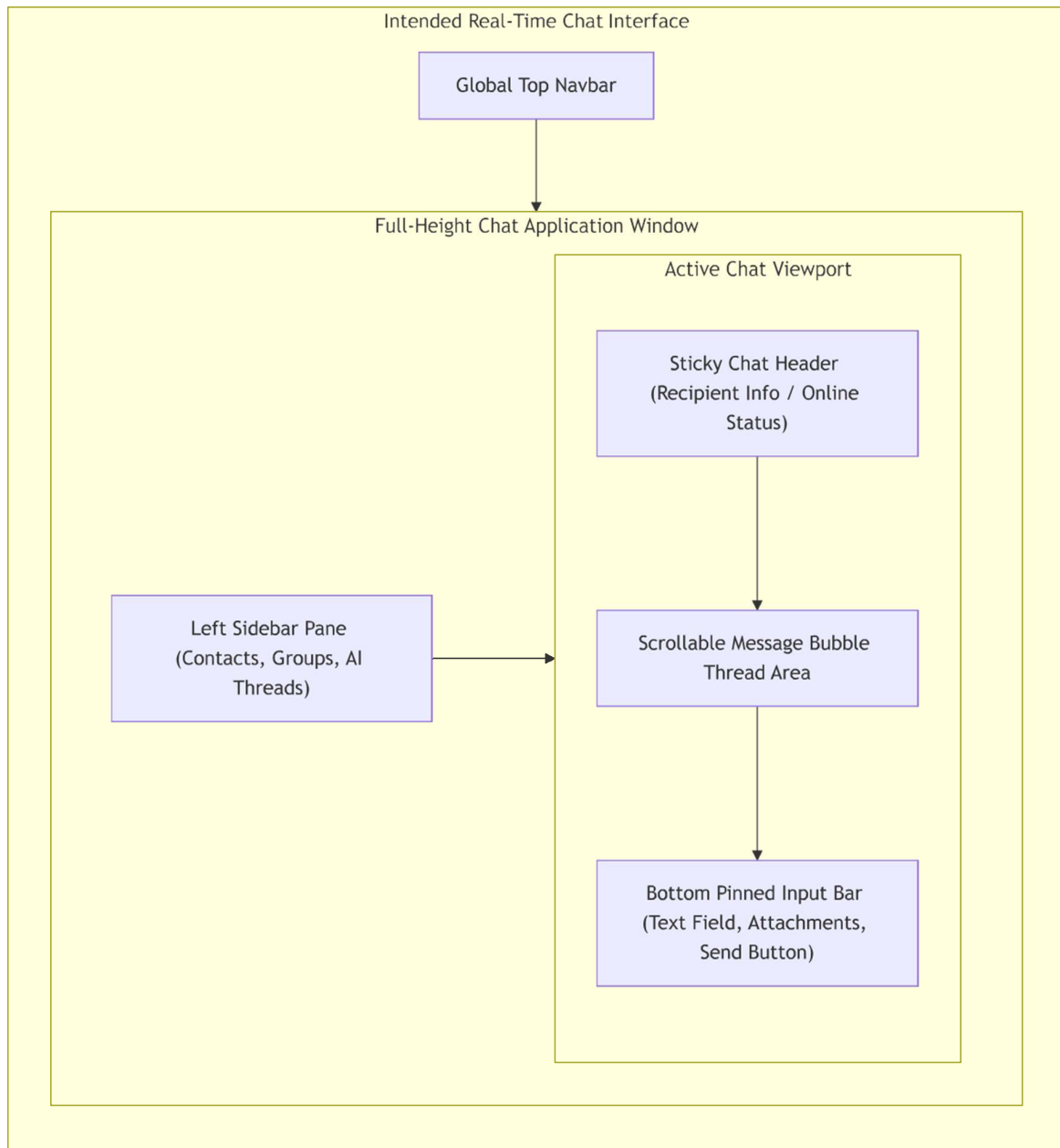
- **Typography:** Core fonts will include 'Inter' for UI elements and 'Roboto' for large text bodies, ensuring high legibility across device form factors.
- **Color Palette:** Deep Blues will be used for primary interactions, Emerald Greens for successful operations, and Rose Red for destructive or error-state actions. The dark mode theme will utilize deep slate gradients to significantly reduce eye strain.
- **Component Library:** We will construct custom-built Tailwind components paired directly with Lucide React to maintain consistent SVG iconography across the portal.

4.2 Major Layout Diagrams (Structural Wireframes)

Below are the structural wireframe blueprints depicting how the primary page layouts will be divided and nested within the Document Object Model (DOM).

Proposed Dashboard Layout Structure





Planned Frontend Component / Page Name	Primary Intended Purpose
About	Will render the core About context. Needs to be integrated into the global router and tested across all mobile breakpoints
ChatPage	Will render the core ChatPage context. Needs to be integrated into the global router and tested across all mobile breakpoints
ContactUs	Will render the core ContactUs context. Needs to be integrated into the global router and tested across all mobile breakpoints
CookiePolicy	Will render the core CookiePolicy context. Needs to be integrated into the global router and tested across all mobile breakpoints
DocumentLab	Will render the core DocumentLab context. Needs to be integrated into the global router and tested across all mobile breakpoints
FriendsPage	Will render the core FriendsPage context. Needs to be integrated into the global router and tested across all mobile breakpoints
Home	Will render the core Home context. Needs to be integrated into the global router and tested across all mobile breakpoints
PrivacyProtocol	Will render the core PrivacyProtocol context. Needs to be integrated into the global router and tested across all mobile breakpoints
PublicOrderTrack	Will render the core PublicOrderTrack context. Needs to be integrated into the global router and tested across all mobile breakpoints
SplitPayPage	Will render the core SplitPayPage context. Needs to be integrated into the global router and tested across all mobile breakpoints
TechnicalHelp	Will render the core TechnicalHelp context. Needs to be integrated into the global router and tested across all mobile breakpoints
TermsOfService	Will render the core TermsOfService context. Needs to be integrated into the global router and tested across all mobile breakpoints
Track	Will render the core Track context. Needs to be integrated into the global router and tested across all mobile breakpoints
WalletPage	Will render the core WalletPage context. Needs to be integrated into the global router and tested across all mobile breakpoints
WalletTopup	Will render the core WalletTopup context. Needs to be integrated into the global router and tested across all mobile breakpoints

5. Proposed Testing & Quality Assurance Plan

Software quality will be maintained through rigorous testing methodologies designed into the CI/CD pipeline right from the inception of development.

Unit Testing: Backend controllers and critical utility functions (such as wallet deduction logic and mathematics for bill splitting) will be unit tested using Jest/Mocha to identify failures in edge cases (e.g., negative inputs, zero values).

Integration Testing: All API endpoints will be continuously validated using tools like Postman to ensure that the request-response cycles match the API specifications and properly persist data into MongoDB.

End-to-End (E2E) Testing: Once the UI is built, simulated user workflows (e.g., Login → Upload Document → Top-up Wallet → Pay → Track) will be automated to guarantee the integrity of the user journey.

6. Proposed Security Architecture

Security will serve as a fundamental pillar of the Reposys infrastructure, implementing multiple layers of defense to protect user data and financial logs.

- **Data in Transit:** 100% of data payloads will be encrypted via TLS/SSL, strictly enforcing HTTPS across all public and internal endpoints.
- **Data at Rest:** Passwords will be cryptographically hashed using bcrypt with dynamic salt rounds before touching the database. Environment variables will be used exclusively to store third-party API secrets.
- **XSS Protection:** The React framework will inherently sanitize inputs and escape string injections prior to DOM rendering, preventing script executions.
- **Rate Limiting:** A robust Rate Limiting middleware will be designed to prevent DDoS attacks and brute-force login attempts on critical authentication routes.

7. Conclusion & Future Scope

7.1 Design Conclusion

The proposed Reposys project architecture successfully conceptualizes a comprehensive digital ecosystem required for modern educational campuses. By tightly defining the schemas, API routes, and interaction models for financial capabilities (Digital Wallet), operational services (Document Printing), and robust communication channels (Real-time Chat, AI Assistant), this design sets a clear and secure path for the development phase.

7.2 Identified System Limitations

Even before development begins, the architecture highlights the following limitations that stakeholders must accept:

- **Network Dependency:** The web-based system will rely heavily on continuous internet connectivity; offline capabilities via Service Workers will be restricted solely to static asset caching.
- **File Size Constraints:** Document uploads for printing will need to be intentionally capped (e.g., 5-10MB limits) to prevent memory exhaustion and buffer overflow on the Node.js application server.

7.3 Future Scope Post-Deployment

Once the initial architecture defined in this document is realized and deployed, future versions of the system may explore the following avenues:

- **Native Mobile Application:** Migrating the React web application footprint to React Native to deliver dedicated iOS and Android applications with integrated biometric authentication.
- **Advanced LLM Grading:** Extending the AI capabilities beyond user support to incorporate automated assignment grading and deep-text summarization features.